

OPERATING SYSTEMS

WEEK- 2 Practical

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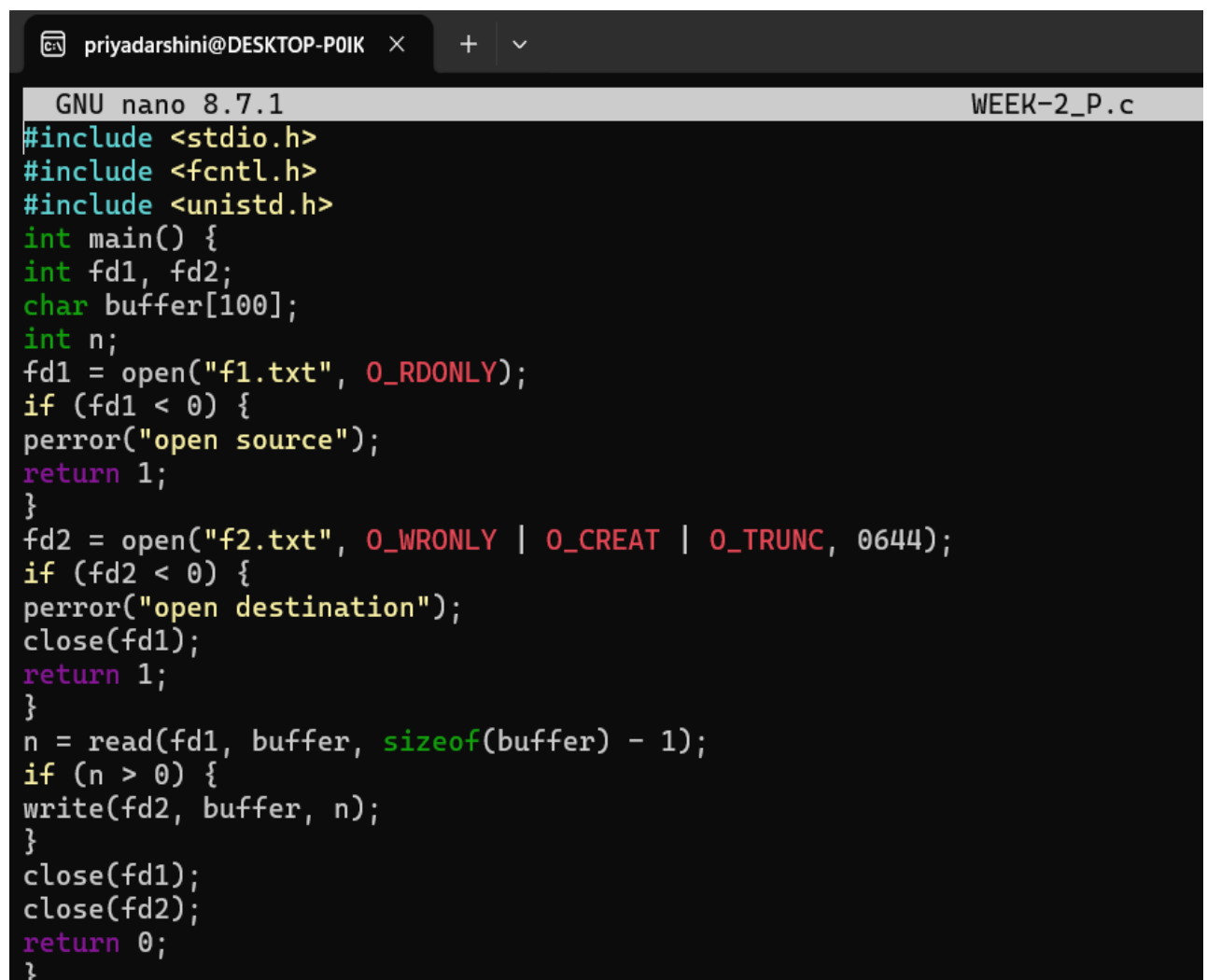
Roll no:- 2520030067

Section no:- 9

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

Program:-



```
priyadarshini@DESKTOP-P0IK  x  +  v
GNU nano 8.7.1 WEEK-2_P.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main() {
int fd1, fd2;
char buffer[100];
int n;
fd1 = open("f1.txt", O_RDONLY);
if (fd1 < 0) {
perror("open source");
return 1;
}
fd2 = open("f2.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
if (fd2 < 0) {
perror("open destination");
close(fd1);
return 1;
}
n = read(fd1, buffer, sizeof(buffer) - 1);
if (n > 0) {
write(fd2, buffer, n);
}
close(fd1);
close(fd2);
return 0;
}
```

Output:-

```
priyadarshini@DESKTOP-P0IK742:~$ nano WEEK-2_P.c
priyadarshini@DESKTOP-P0IK742:~$ gcc WEEK-2_P.c
priyadarshini@DESKTOP-P0IK742:~$ ./a.out
priyadarshini@DESKTOP-P0IK742:~$ cat f2.txt
Hello, this is my test file.
Copy me into f2.txt.
```